Evidently, in this example, the inclusive **or** is not symmetric. The truth of (*A* **or** *B*) may depend on the order in which we confirm the two propositions. This is not a small thing; it means not only that the laws of quantum physics are different from their classical counterparts, but that the very foundations of logic are different in quantum physics as well.

What about (*A* **and** *B*)? Suppose our first measurement yields $\sigma_z = +1$ and the second, $\sigma_x = +1$. This is of course a possible outcome. We would be inclined to say that (*A* **and** *B*) is true. But in science, especially in physics, the truth of a proposition implies that the proposition can be verified by subsequent observation. In classical physics, the gentleness of observations implies that subsequent experiments are unaffected and will confirm an earlier experiment. A coin that turns up Heads will not be flipped to Tails by the act of observing it—at least not classically. Quantum mechanically, the second measurement ($\sigma_x = +1$) ruins the possibility of verifying the first. Once σ_x has been prepared along the *x* axis, another mesurement of σ_z will give a random answer. Thus (*A* **and** *B*) is not confirmable: the second piece of the experiment interferes with the possibility of confirming the first piece.

If you know a bit about quantum mechanics, you probably recognize that we are talking about the uncertainty principle. The uncertainty principle doesn't apply only to position and momentum (or velocity); it applies to many pairs of measurable quantities. In the case of the spin, it applies to propositions involving two different components of σ. In the case of position and momentum, the two propositions we might consider are:

> A certain particle has position *x*.
>
> That same particle has momentum *p*.

From these, we can form the two composite propositions

> The particle has position *x* **and** the particle has momentum *p*.
>
> The particle has position *x* **or** the particle has momentum *p*.

Awkward as they are, both of these propositions have meaning in the English language, and in classical physics as well. However, in quantum